

Exercise - Week 2: MATH1061/7861

Duration: 30 minutes

2 Questions - 10 marks

Topics: Logical Forms and Logical Equivalences

Student Name:

Student number: |—|—|—|—|—|—|—|—|

Your student number should be 8 digits and **not** start with an s. e.g. 41234567

- (1) **(5 marks)** Let p, q, r be statement variables.
Consider the following three statements:

$$\begin{aligned}S_1 &= p \vee \sim q \\S_2 &= (p \wedge r) \vee q \\S_3 &= \sim p \vee (q \wedge r)\end{aligned}$$

Task: Using the truth table below find all triples (p, q, r) such that the three statements S_1 , S_2 , and S_3 have the same truth value.

p	q	r	$\sim p$	$p \wedge r$	$q \wedge r$	$p \vee \sim q$	$(p \wedge r) \vee q$	$\sim p \vee (q \wedge r)$	Equal?
T	T	T							
T	T	F							
T	F	T							
T	F	F							
F	T	T							
F	T	F							
F	F	T							
F	F	F							

- (2) **(5 marks)** For each of the following, either prove that the statements are logically equivalent using the laws of logical equivalence, or give values which show that they are not equivalent.

Label the laws of logical equivalence used in each step.

(a) $(p \vee \sim q) \vee (\sim p \wedge \sim q) \equiv p$

(b) $\sim (p \vee \sim q) \vee (\sim p \wedge \sim q) \equiv \sim p$

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Logical Equivalences

Given any statement variables p , q and r , a tautology $\mathbf{t}$ and a contradiction $\mathbf{c}$, the following logical equivalences hold.

<i>Commutative laws</i>	$p \wedge q \equiv q \wedge p$	$p \vee q \equiv q \vee p$
<i>Associative laws</i>	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	$(p \vee q) \vee r \equiv p \vee (q \vee r)$
<i>Distributive laws</i>	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
<i>Identity laws</i>	$p \wedge \mathbf{t} \equiv p$	$p \vee \mathbf{c} \equiv p$
<i>Negation laws</i>	$p \vee \sim p \equiv \mathbf{t}$	$p \wedge \sim p \equiv \mathbf{c}$
<i>Double negative laws</i>	$\sim(\sim p) \equiv p$	
<i>Idempotent laws</i>	$p \wedge p \equiv p$	$p \vee p \equiv p$
<i>Universal bound laws</i>	$p \vee \mathbf{t} \equiv \mathbf{t}$	$p \wedge \mathbf{c} \equiv \mathbf{c}$
<i>De Morgan's laws</i>	$\sim(p \wedge q) \equiv \sim p \vee \sim q$	$\sim(p \vee q) \equiv \sim p \wedge \sim q$
<i>Absorption laws</i>	$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$
<i>Negations of $\mathbf{t}$ and $\mathbf{c}$</i>	$\sim \mathbf{t} \equiv \mathbf{c}$	$\sim \mathbf{c} \equiv \mathbf{t}$